STAFF REVIEW · H446 131 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Securely sharing and managing the secret key.

2. An input used by a subroutine.

3. Sum 0, carry-out 1.

4. 24.

I Do — source worked steps

1. Bob shares public key — Predict the first step from the prompt.

2. Alice encrypts for Bob — Explain the intermediate result or decision.

3. Bob uses private key — Alice encrypts a suitably sized message or session key using Bob's public key. Bob decrypts using his matching private key. Alice never requires Bob's private key.

We Do — guided CHECK

1. Publisher signs privately
Predict the first step from the prompt.
Distribute signed file
Explain the intermediate result or decision.
Verify with public key
The publisher creates the digital signature with its private key. Users verify it with the public key obtained through a trusted route; this supports authenticity and integrity, not secrecy of the update.

You Do — source mark schemes

1. Public key distributed [2], private key secret [2], public encrypt/private decrypt for recipient [2].

2. Signer signs using private key [2], verifier uses trusted public key [2], checks authenticity/integrity [2].

3. Symmetric efficient for bulk [2], asymmetric useful for secure session-key exchange [2], hybrid approach [2], key identity/trust and limitations [2].

Exit Ticket — source answer

1. False. The matching private key must be kept secret.